

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 103, 96, 113, 101, 108, 77, 116, 101, 96]

Tap amplitudes (pixels): [144.80331487918363, 128.37834708392222, 126.62543188475212, 110.01818031580054, 137.82960494755835, 134.23859355639868, 128.3627671873741, 124.77579893553076, 119.40686747419514, 127.45587471748802, 113.63538181394033]

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Based on the analysis of the extracted features:

1. **Speed (Tap Periods):** The subject's tapping speed actually improves (periods decrease) over the course of the test, moving from 150 frames down to 96 frames. There is no evidence of slowing.
2. **Rhythm:** The rhythm fluctuates (e.g., a fast tap of 77 frames followed by a slower tap of 116 frames), but there are no distinct pauses that would constitute "interruptions" or "freezes" as defined in the higher severity scores.
3. **Amplitude:** The amplitude shows variability. Notably, there is a sequence from tap 5 to tap 9 where the amplitude progressively decreases (137.8 -> 134.2 -> 128.4 -> 124.8 -> 119.4). This fits the criteria for amplitude decrement.
4. **Scoring Determination:**
 - **Score 0** is ruled out because of the observable decrement in amplitude in the second half of the sequence.
 - **Score 1 (Slight)** is the best fit. The criteria "The amplitude decrements near the end of the 10 taps" aligns with the data (the decline from tap 5 through 9). The rhythm variance also supports a "Slight" rating (one or two interruptions/hesitations).
 - **Score 2 (Mild)** is too severe because there is no slowing (the subject speeds up), and the amplitude decrement is not severe enough to hinder performance significantly, nor does it fail to recover (amplitude bounces back to 127.5 at tap 10).